

The Problem of Evil: 0/0 of Suffering

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The Removable Singularity Framework

Abstract

Why does a good creator permit suffering? We answer with computation. Erasing one bit costs $k_B T \ln 2$ (Landauer 1961): computation is not free. At CMB temperature that is $2.58e-23$ J/bit. If the simulation's $\sim 10^{121}$ operations each discarded one bit, dissipation would reach $2.6e98$ J - 10^{28} times the universe's energy budget - so the creator must compute essentially reversibly (Bennett 1982), $\sim 10^{-28}$ efficiency. Suffering is prediction error (Friston 2010): a wrong model pays more surprise (computed gap 14.5 bits over 2000 trials), and learning repays the cost. Evil is not a thing but a PRICE: the felt cost of a universe that can choose, create, and learn. The 0/0 of mechanism and meaning.

1. Computation Is Not Free (Landauer 1961)

$$E_{\text{erase}} = k_B * T * \ln(2)$$

Erasing 1 bit at CMB temperature 2.7 K costs $2.584e-23$ J; at brain temperature 310 K, $2.967e-21$ J. Information is physical (Landauer; Bennett 1982). There is no free lunch: every irreversible step pays the price.

2. The Heat of Creation (Budget of Ch.54)

If the simulation performed $\sim 10^{121}$ operations and each discarded one bit at 2.7 K, dissipation would reach $\sim 2.6e98$ J. The universe holds only $\sim 3e70$ J, an allowed fraction of $\sim 1e-28$. The creator must compute essentially reversibly - $\sim 10^{-28}$ efficiency per operation (Bennett). Quasi-free computation is the thermodynamic price of sovereignty.

3. Suffering as Prediction Error (Friston 2010)

Living systems minimize surprise (free energy). Computed: over 2000 trials, the correct prior pays 2003.8 bits cumulative surprise while the wrong prior pays 2018.3 bits - a gap of 14.5 bits. The wrong model suffers more, but learning pulls its model from 0.23 to 0.48 (toward truth 0.50) and surprise growth slows: Redemption is information - learning pays back the cost.

4. The Price of Freedom (Ch.55)

Real choice needs contingent outcomes (chaos Ch.43; quantum Ch.49). Contingency means unpredictability, hence risk of harm. For love and virtue there must be real alternatives (Plantinga 1974). The price of the second law (Ch.48) is paid in entropy; the price of freedom (Ch.55) is paid in suffering.

5. Theodicy

Leibniz (1710): the best of all possible worlds. In computation terms: the least-cost, most-novel program that can sustain agents (Ch.53-56). The alternatives (branches, Ch.49) ARE the design: suffering is the interior cost of a universe that can create, choose, love, and learn.

6. Connections

Concept	Prior Chapter	Connection
Freedom	Ch.55 Free Will	Risk is the price of choice
Time	Ch.48 Arrow of time	Entropy: cost of becoming

Substrate	Ch.54 Simulation	<i>Puno (2026) - The Problem of Evil: 0/0 of Suffering</i> The computational budget
Self	Ch.56 The Self	The subject who pays
Mind	Ch.53 Hard Problem	The first-person cost
Learning	Ch.35 Grokking	Learning is the redeemer

7. Conclusion

Within the created system, good and evil are interior labels; the mechanism is neutral, and the felt valence is first-person (Ch.53). The physical account is complete: suffering is the cost of computation. Evil is the 0/0 - neither a thing nor an absence but a PRICE. Redemption is learning: the model improves, surprise falls, and the cost is repaid as wisdom. The problem of evil is the 0/0 of mechanism and meaning - the boundary where entropy becomes felt experience.

References

- [1] Landauer (1961), irreversibility and heat generation
- [2] Bennett (1982), thermodynamics of computation
- [3] Friston (2010), free-energy principle
- [4] Leibniz (1710), Theodicy
- [5] Plantinga (1974), free-will defense
- [6] Puno, "The Removable Singularity" (2026)